



From Accounts to Alerts: EDA and Logistic Regression for Early Corporate Risk Prediction

Yun-Hsuan Chang, Chia-Yen Chang, Huei-Wen Teng



Outline

1. Motivation
2. Data & EDA
3. Research Design
4. Model Evaluation (AUC, Calibration, Brier Score)
5. Conclusion & Implications



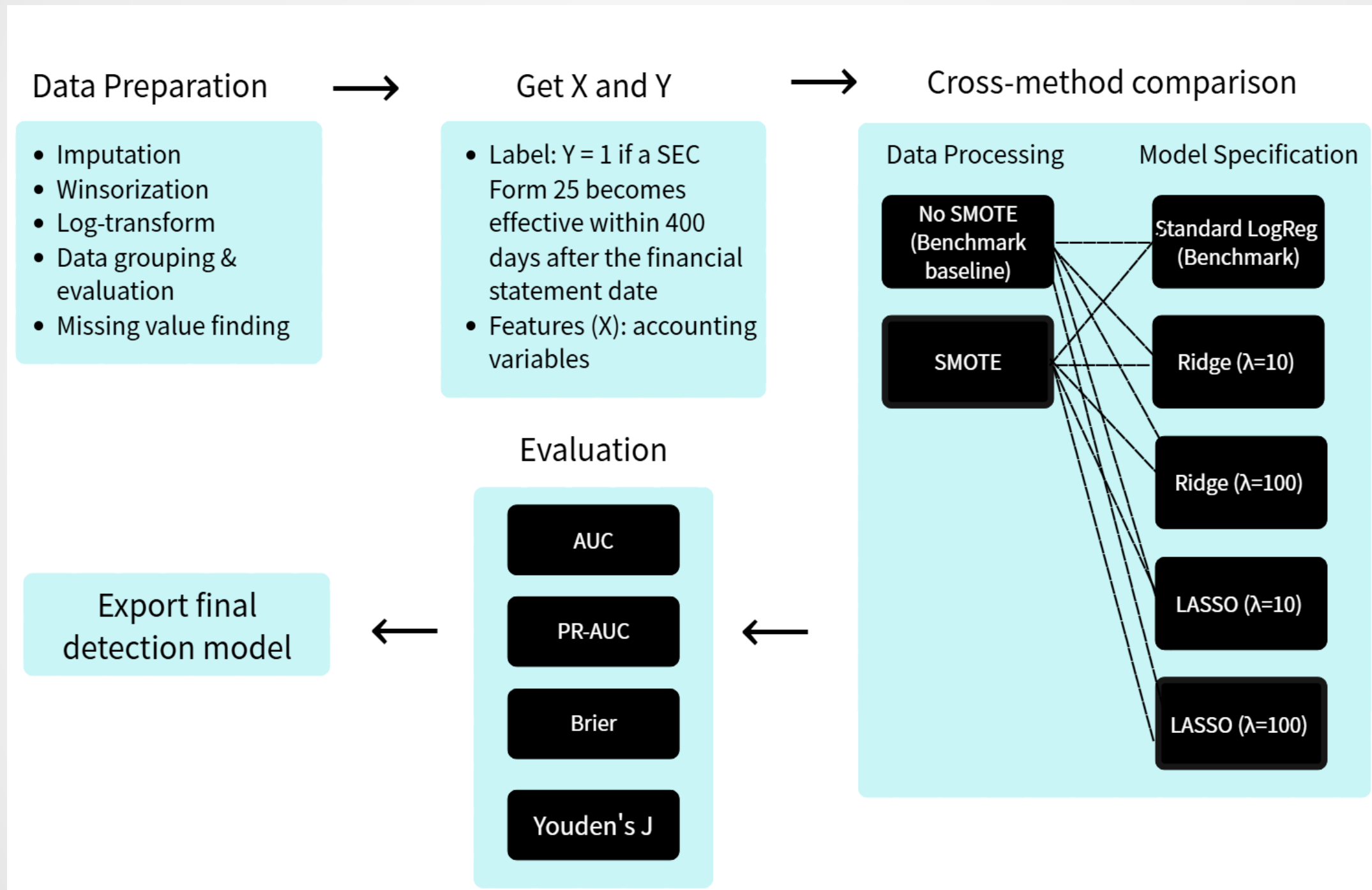
Motivation

Fast-moving corporate world, risks can emerge quietly.
If we make decision from the news, it's too late.

- Early Warning Systems - not just useful, but essential.
 - Accessible – inputs are openly available
 - Trustworthy – the model can be explained clearly
 - Actionable – early alerts give time to respond
- Transparency and Interpretability
- Publicly Available Data



Graphical Abstracts



Overview

- USA annual financial statements (2019–2024) + SEC Form 25 delisting labels
- Label: $Y=1$ if a Form 25 becomes effective within 400 days after the financial statement date
- Train: 2019–2023, Test: 2024



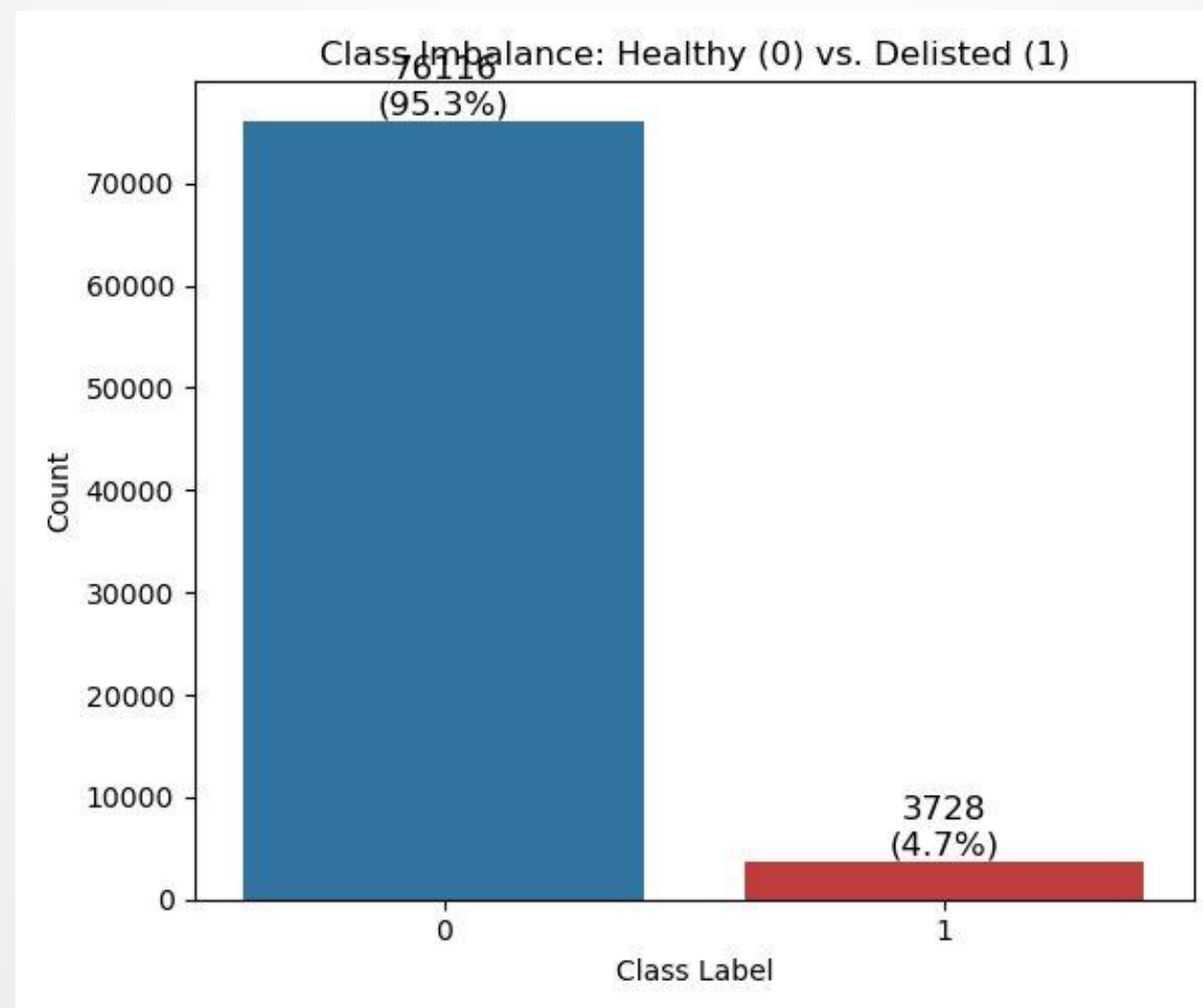
Feature Set Overview

- ▣ Metadata / Identifiers (drop) : costat, curcd, datafmt, indfmt, consol, gvkey, datadate, cik, sic, fyear
- ▣ Liquidity: act, lct, rect, invt, che
- ▣ Leverage: dlc, dltd, lt, seq
- ▣ Profitability: sale, cogs, ebit, oibdp, ni, dp
- ▣ Cash Flow: Oancf
- ▣ Market / Equity: csho, prcc_f
- ▣ Expenses / Other Accounting Items: txt, xint, xsga
- ▣ Target Variable: $Y=1$ if Form 25 effective within 400 days



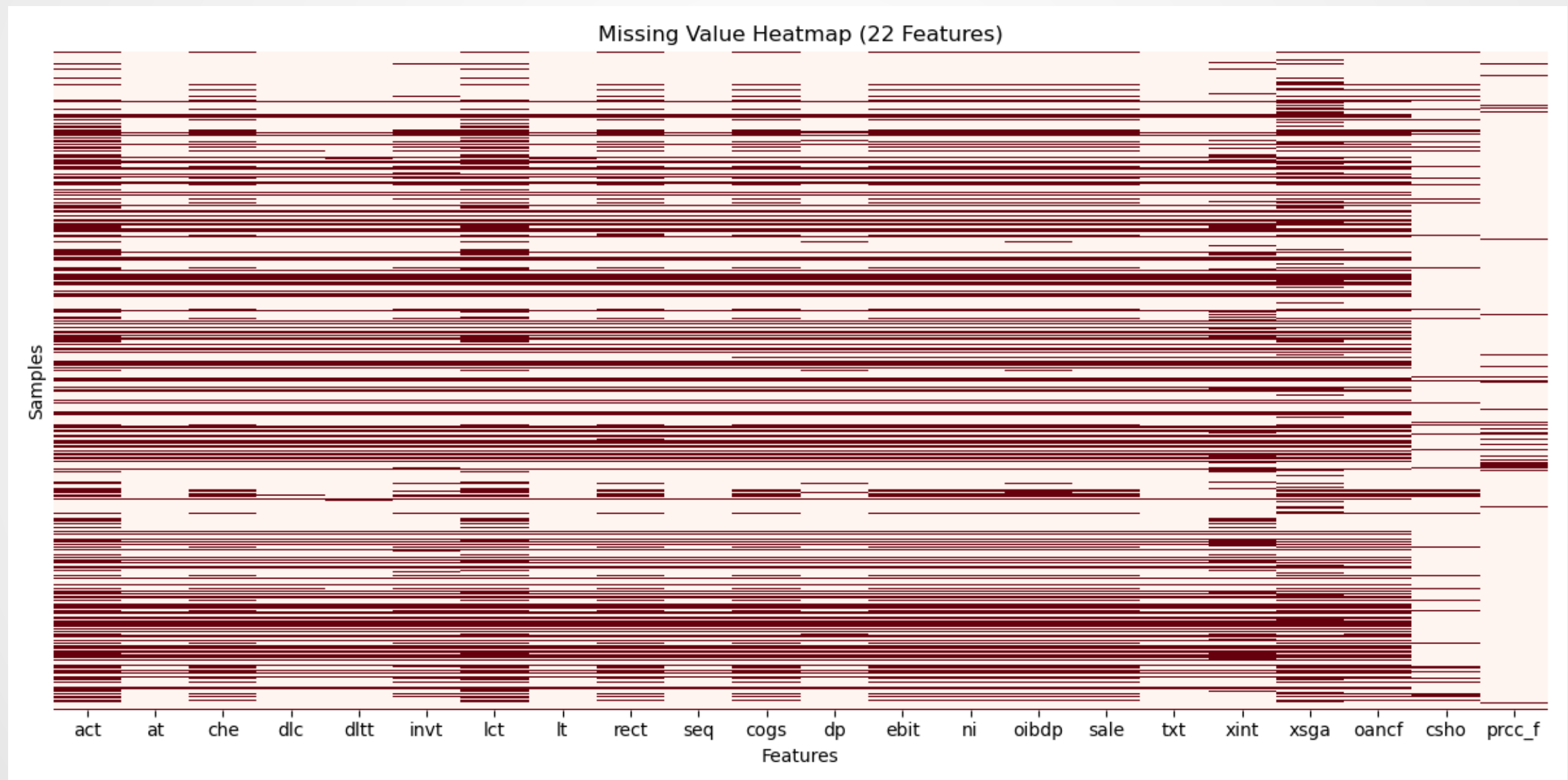
Data Overview

- ▣ **Source:** Fundamentals Annual (US Companies).
- ▣ **Scale:** 16237 Companies / 79844 Records.
- ▣ **Target (Y=1):** Form 25 filing within 400 days.
- ▣ **Imbalance:** Default rate is only ~4.7% (Highly Imbalanced).



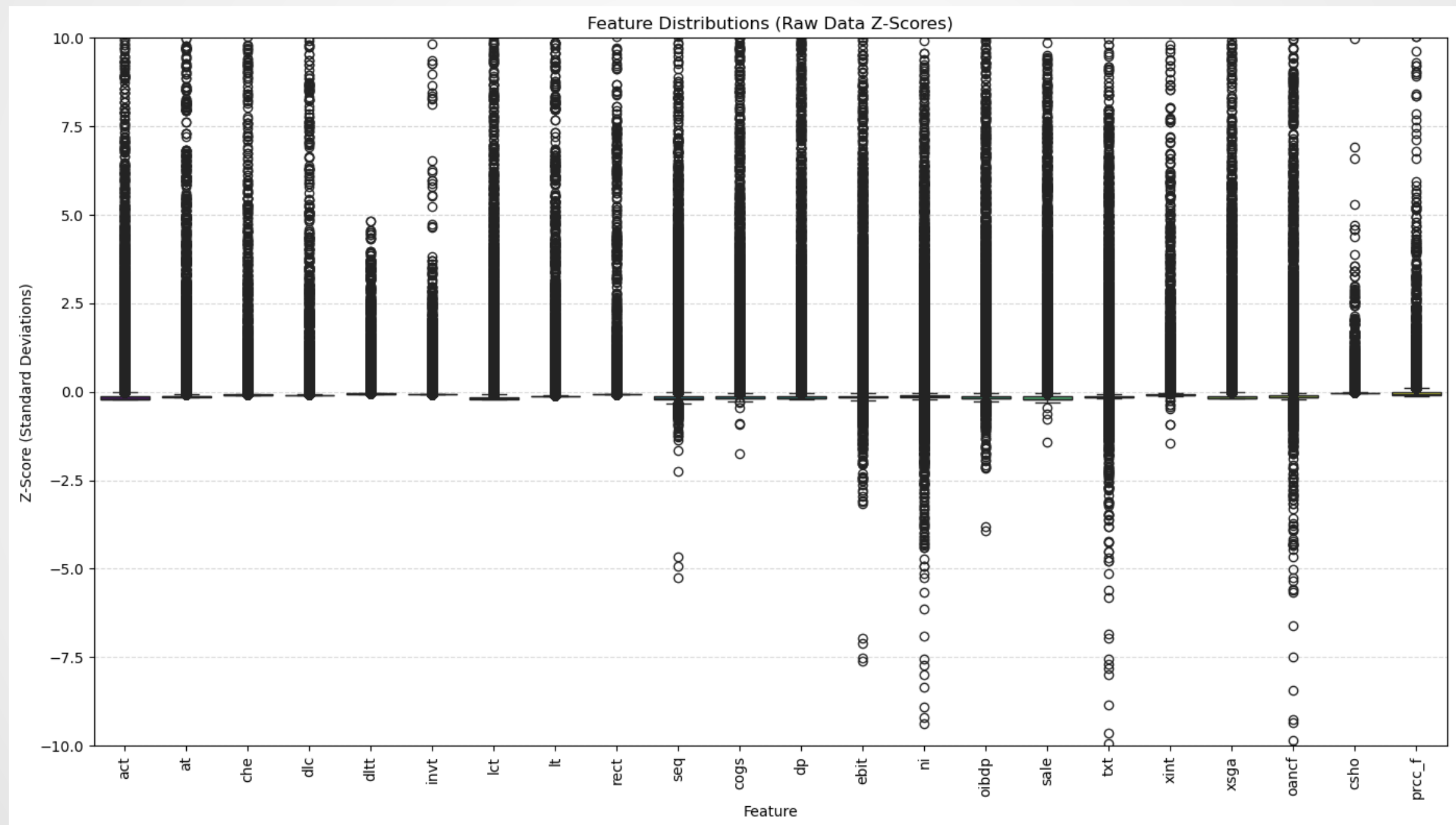
Data Quality & Missingness

- Average missing rate across all features: 36.16%
- Impute: SIC2×year median (train-only); else overall train median



Data

- heavy-tailed and asymmetric distributions



Data

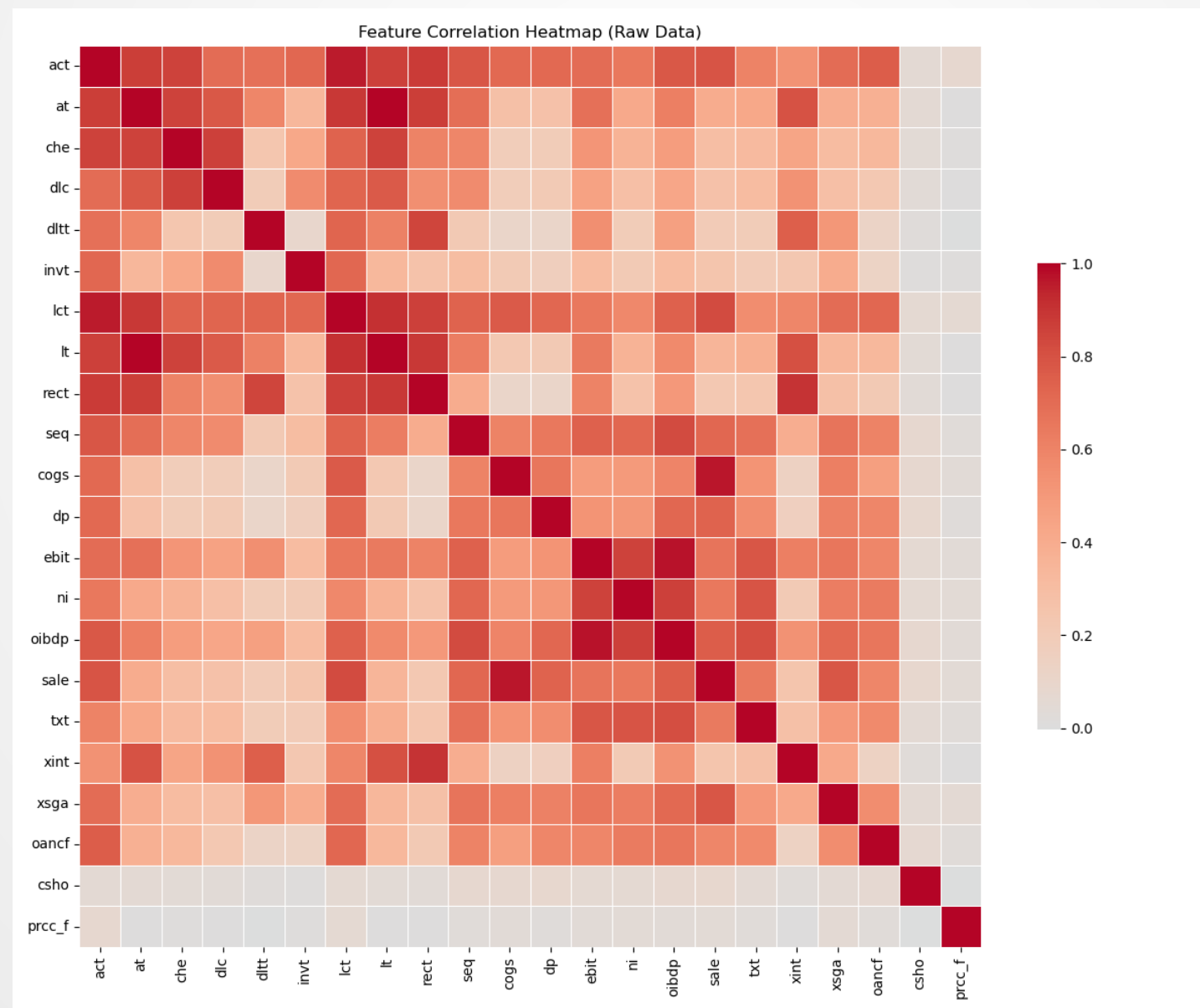
- Most variables are highly skewed
- Several variables exhibit extreme kurtosis (>10)
- apply winsorization and log-transform before model estimation

	count	mean	std	min	50%	max	skewness	kurtosis
act	40613	2009.53	9346.14	0	138.58	264889.4	12.17	202.25
at	54110	24980.98	174056.8	0	799.87	4349731	13.92	234.3
che	47968	2323.78	26041.07	-0.01	57.96	1058786	23.02	630.73
dlc	53568	1561.54	16732.51	0	8	614237.4	19.72	463.65
dltt	54002	4781.37	75563.64	0	60.48	4216909	46.49	2319
inv	49218	663.86	9357.72	0	0.55	530917	36.6	1601.39
lct	40651	1642.82	8149.56	0	58.17	227561.9	12.02	198.57
lt	54036	21403.05	163212.2	0	429.85	4255074	14.43	251.12
rect	47331	5607.91	71936.99	-0.07	37.7	4150750	32.91	1495.26
seq	54110	3396.04	16038.85	-80708	218.77	649368	14.67	344.73
cogs	47805	2781.34	14751.12	-23105.2	83.74	478289	14.08	270.34
dp	51688	255.35	1282.09	-4.12	6.52	52892	14.16	289.58
ebit	47805	625.69	3814.52	-28387	4.46	141290	17	414.34
ni	47805	339.16	2472.64	-22819	0.13	100118	19.35	585.03
oibdp	46470	882.83	4588.87	-17188	11.92	141290	13.85	276.05
sale	47806	4284.76	20369.63	-24954.7	173.62	645737	13.2	249.14
txt	53927	108.31	734.07	-8518	0.64	49860	20.46	759.49
xint	47796	270.08	2761.12	-3775	7.15	121818	25.96	808.96
xsga	40251	740.92	4031.53	-12.09	52.28	242315	24.62	1044.99
oancf	47672	698.61	4575.3	-100815	9.1	231786	15.94	484.49
csho	72026	247.51	6108.91	0	29.83	606407.7	78.48	6630.78
prcc_f	74655	48.24	418.38	0	20.33	31700	51.67	3098.36



Data

- High correlation exists between financial features (e.g., Assets vs. Liabilities).
- the heatmap reveals strong multicollinearity



Benchmark Model

- Benchmark Model: Standard Logistic Regression (No Penalty).
- Data: Raw Data with Standard Imputation.
- Training Split: 2019-2023 data.
- Testing Split: 2024 data



Research Design (2x5 Matrix):

- ▣ Sampling: No SMOTE vs. SMOTE.
- ▣ Models:
 1. Simple LR
 2. Lasso ($\lambda=10$)
 3. Lasso ($\lambda=100$)
 4. Ridge ($\lambda=10$)
 5. Ridge ($\lambda=100$)

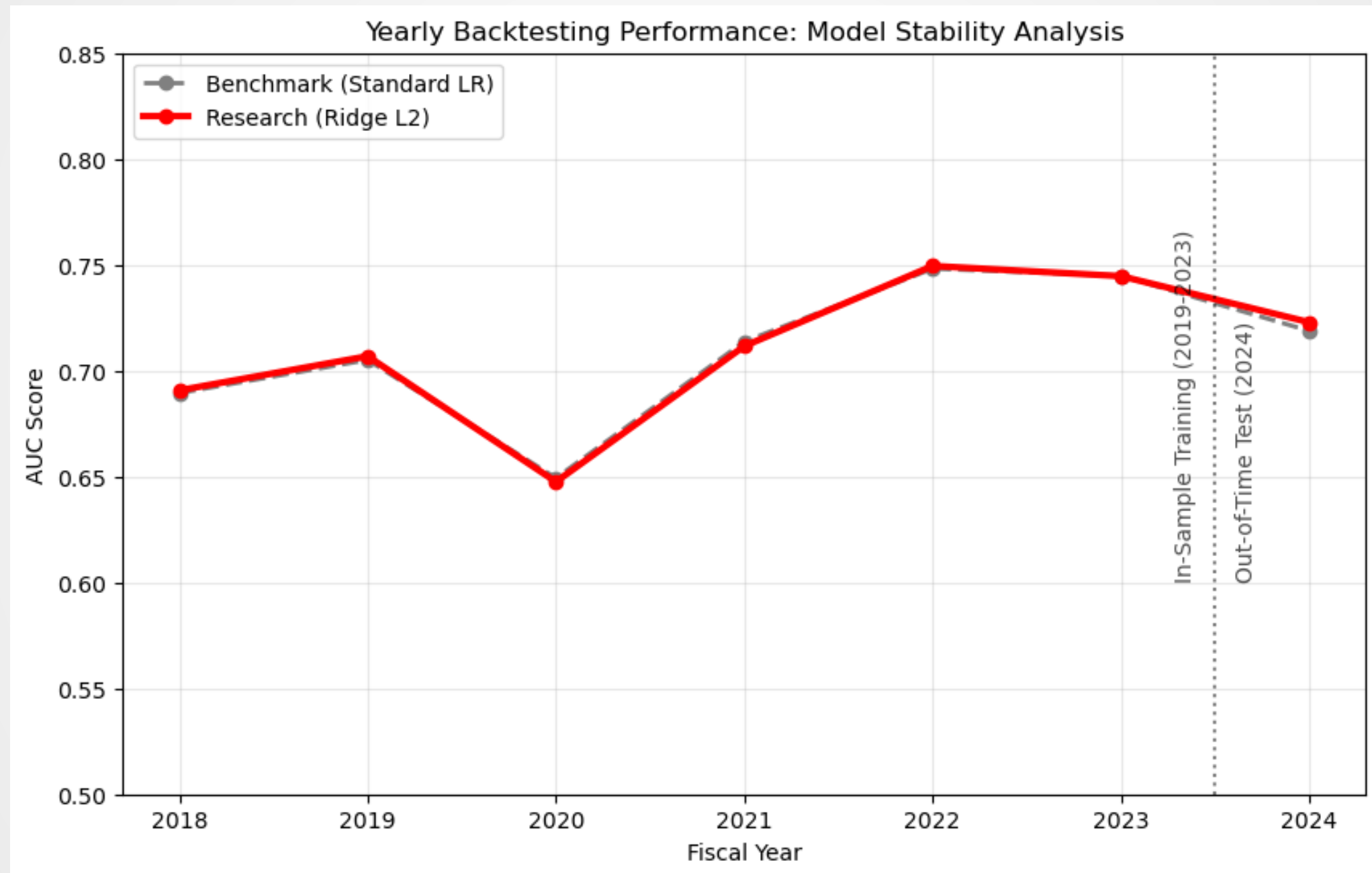


Yearly Backtesting Results

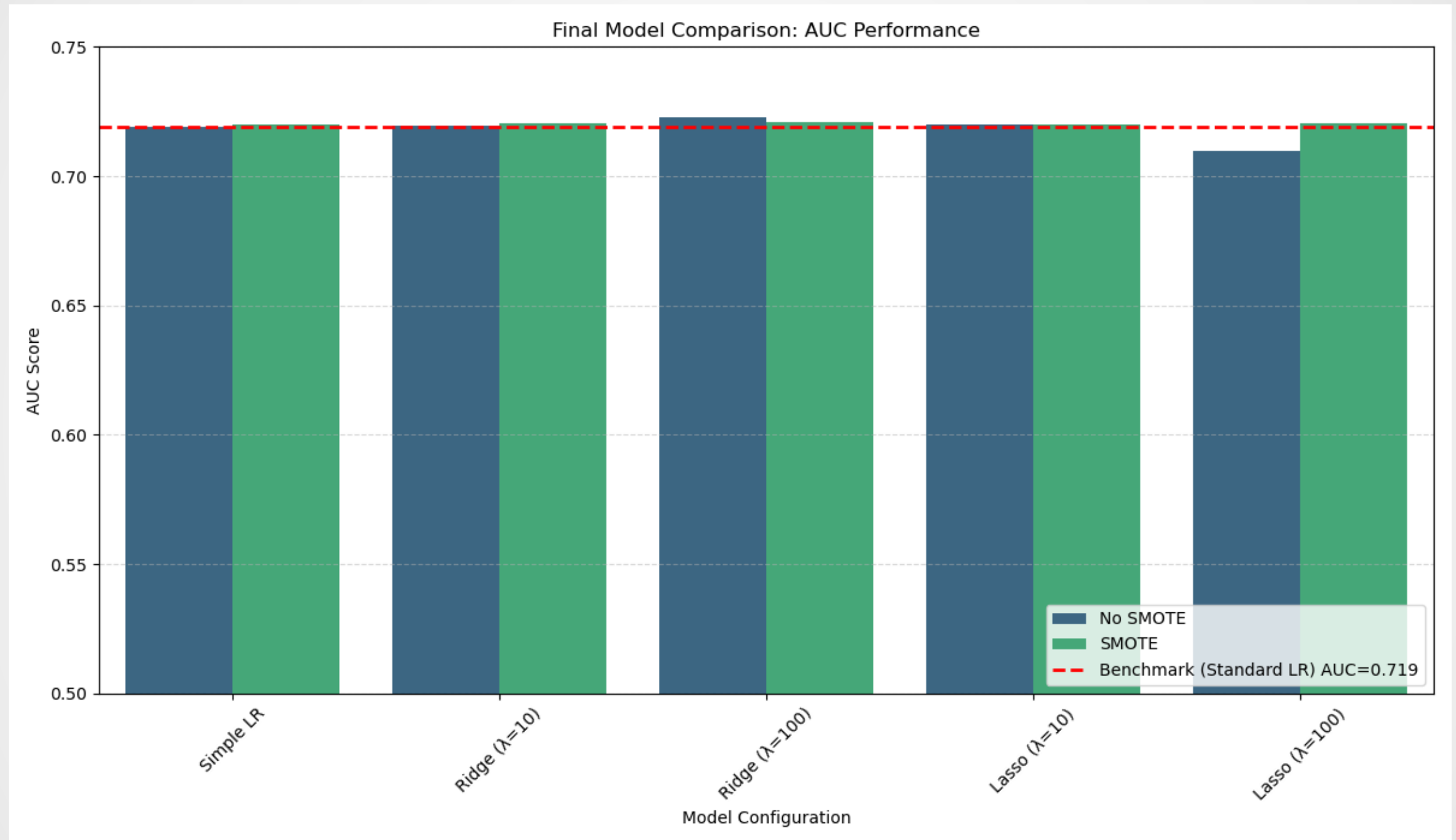
Year	Samples	Defaults	Def_Rate	Benchmark AUC	Research AUC	Type	
2018	944	33	0.034958	0.689452	0.690716	Train (In-Sample)	
2019	12660	551	0.043523	0.705081	0.706846	Train (In-Sample)	
2020	12970	793	0.061141	0.649308	0.647548	Train (In-Sample)	
2021	13312	604	0.045373	0.713732	0.711614	Train (In-Sample)	
2022	13573	664	0.048921	0.748193	0.749468	Train (In-Sample)	
2023	13562	598	0.044094	0.744849	0.744651	Train (In-Sample)	
2024	12823	485	0.037823	0.718823	0.72287	Test (Out-of-Time)	



Yearly Backtesting Results



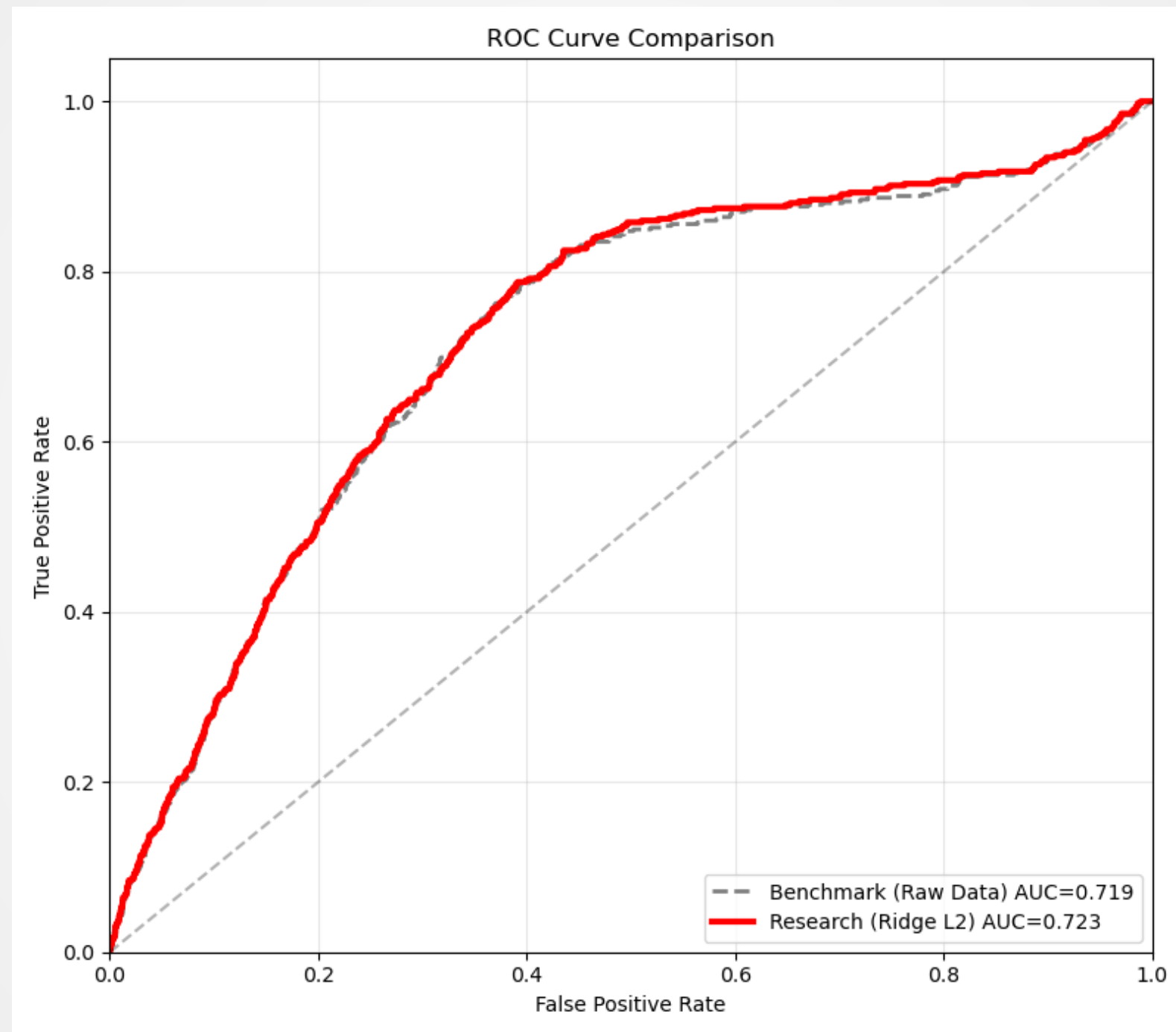
AUC Performance



Ridge > Lasso > Benchmark



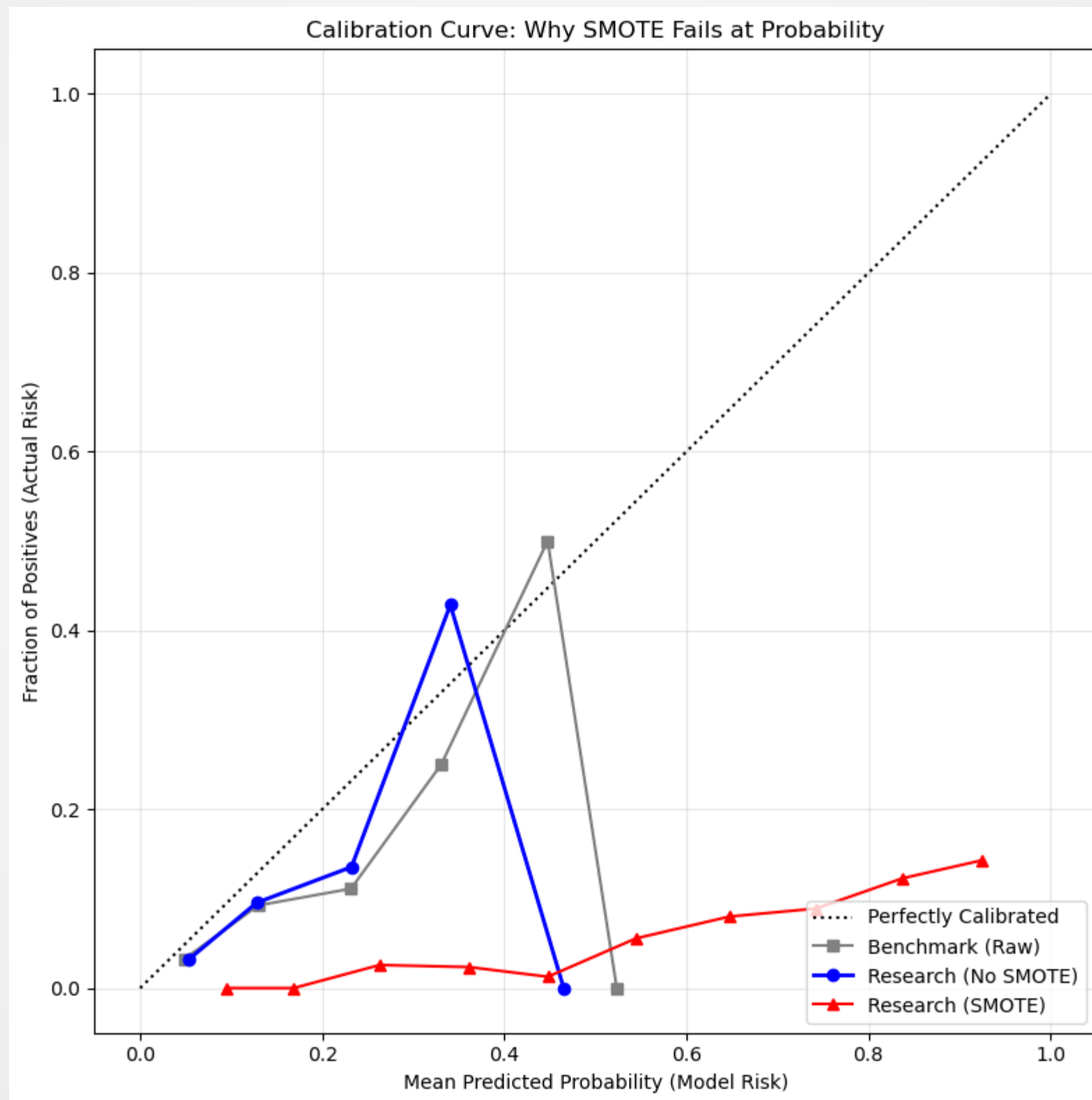
AUC Performance (Benchmark vs The Best Outcome)



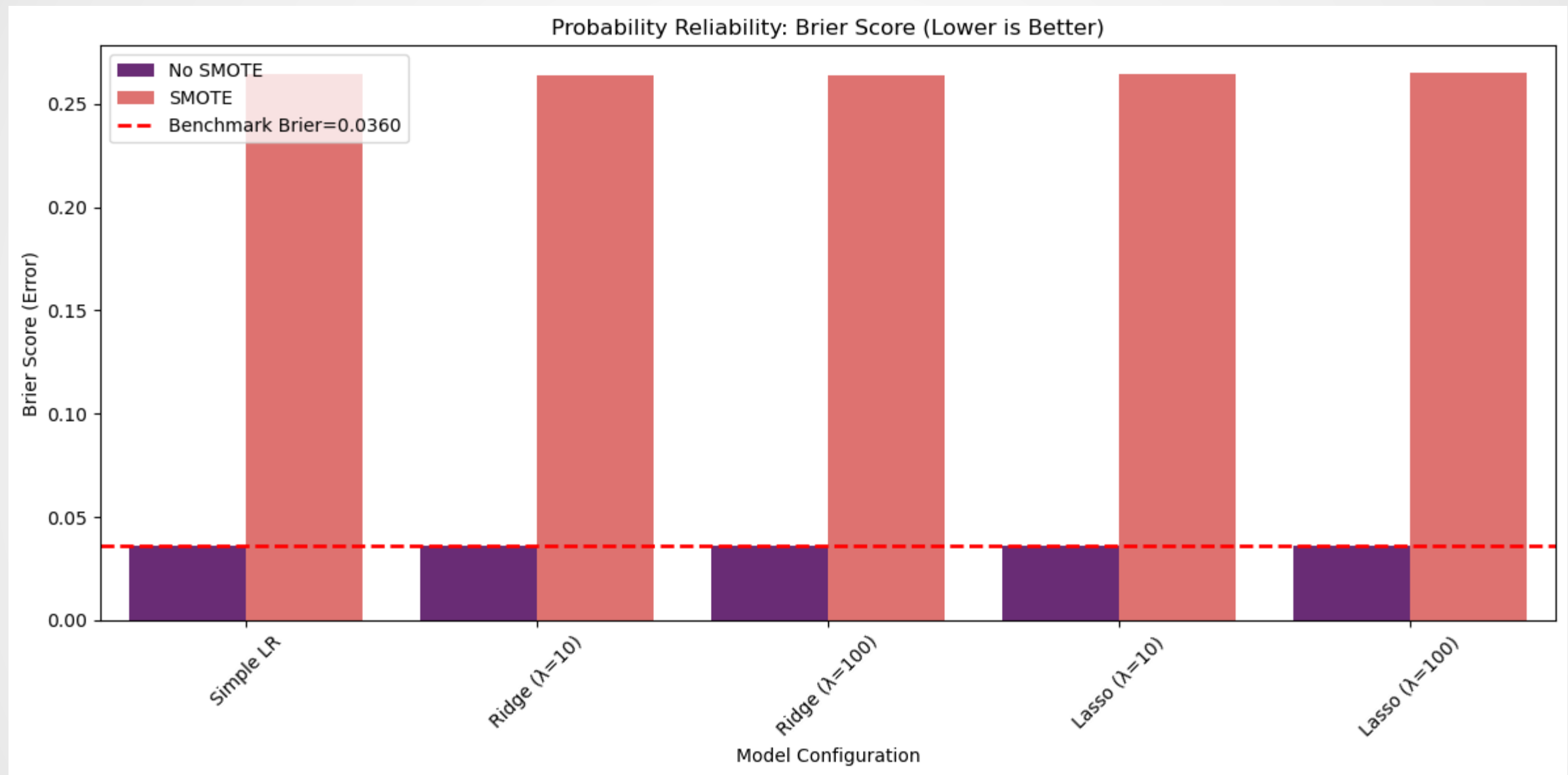
- Benchmark: ~0.718 (Red Line).
- Winner: No SMOTE + Ridge ($\lambda=100$) achieved the highest AUC (~0.723).



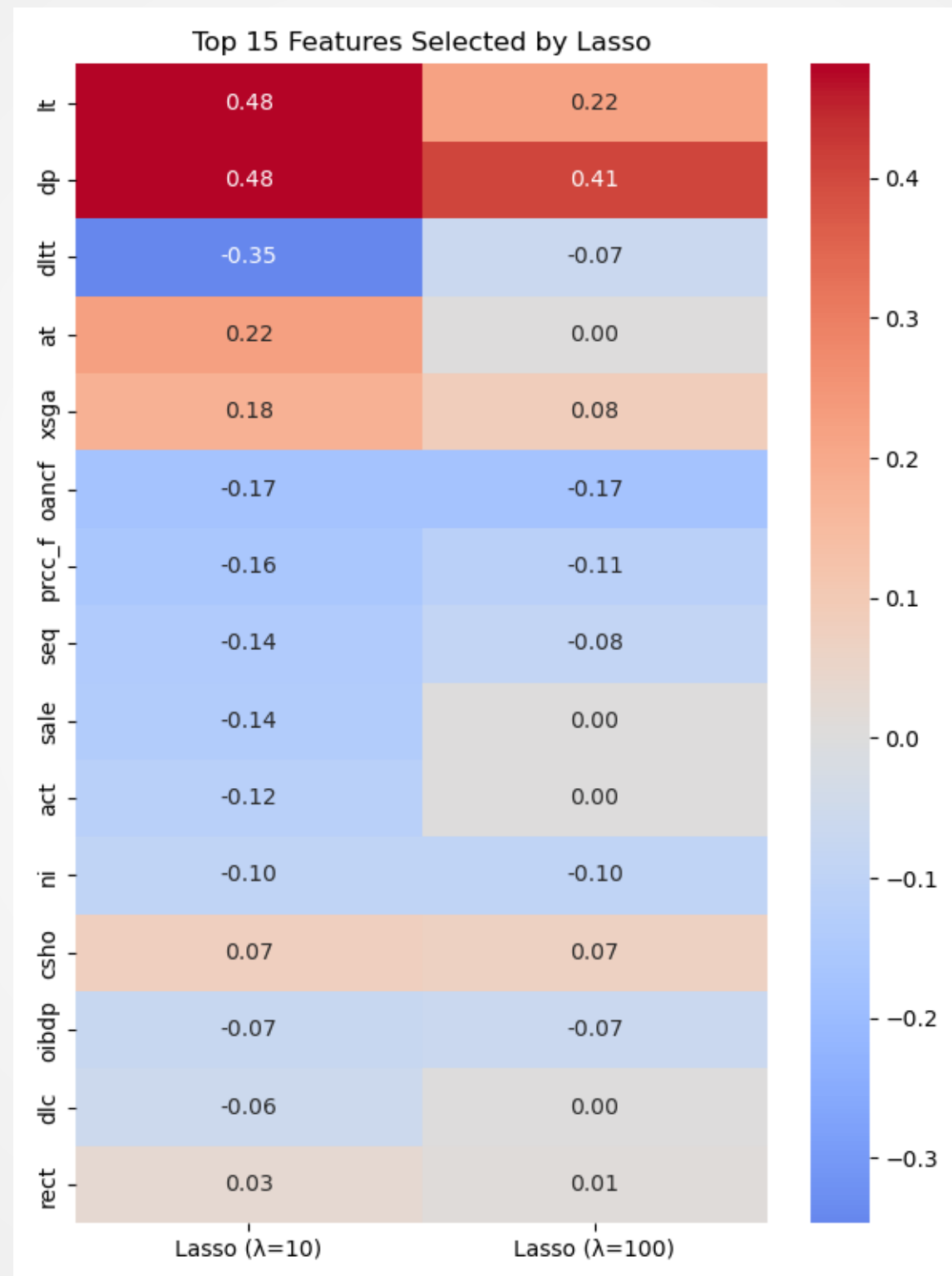
Calibration Curve: Why SMOTE Fails at Probability



Probability Reliability: Brier Score (Lower is Better)



Feature Selection via Lasso



Conclusion

- ▣ **Data Quality Wins:** Industry-specific imputation and outlier removal enabled simple linear models to achieve high performance (**AUC 0.72**).
- ▣ **Regularization Strategy:** Ridge (L2) is superior to Lasso for this dataset, suggesting that handling multicollinearity is more important than feature elimination.
- ▣ **Sampling Caution:** SMOTE degrades probability calibration.
- ▣ **Cost-sensitive learning** (Class Weights) is a better alternative for financial risk models.





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